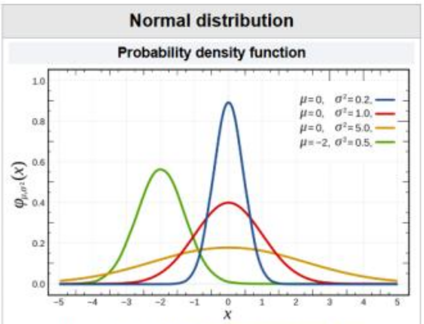
Normal Or Gaussian Distribution

08 June 2026 14:20

Normal distribution:- * ⇒ Imp question



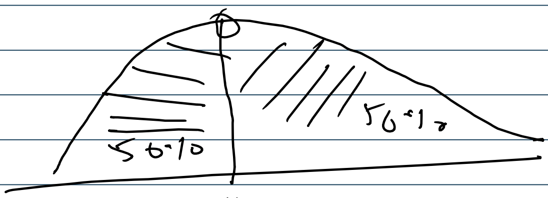
The red curve is the standard normal distribution.

In probability theory and statistics, a **normal distribution** or **Gaussian distribution** is a type of continuous probability distribution for a real-valued random variable.

Anything ⇒ Normal distribution

⇒ discrete ✗ → continuous

we	Age
60.1 kg	30
70.5 kg	40
	50
	60



⇒ Symmetrical distribution *

μ =

Notation = $N(\mu, \sigma^2)$

Mean(μ) = Median = Mode

Parameters

Mean ← $\mu \in R$ = (mean)

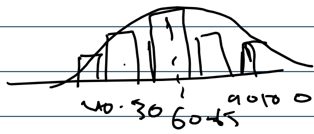
$\sigma^2 \in R > 0$

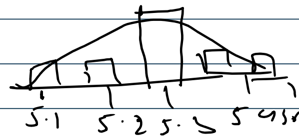
$x \in R$ continuous

60, 65, 56, 60, 65, 60

Ex:- weight of student in a class

Height of stud in a class





$$\boxed{N.D.F = \frac{1}{\sigma\sqrt{2\pi}} e^{-1/2\left[\frac{x_i - \mu}{\sigma}\right]^2}}$$

Mean

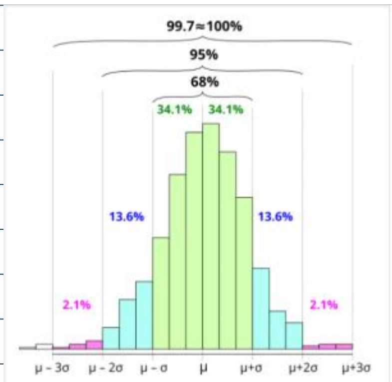
$$\mu = \sum_{i=1}^{n} \frac{x_i}{n}$$

Variance

$$\sigma^2 = \sum_{i=n}^{n} \frac{(x_i - \mu)^2}{n}$$

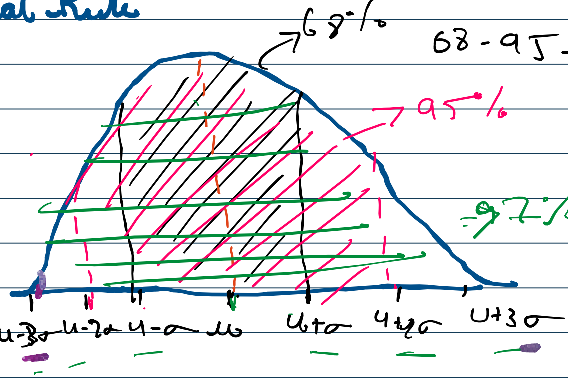
S.d = $\sqrt{\text{Varia}}$

$= \sqrt{\sigma^2}$



Empirical Rule

68-95-99.7%



40 4.245 46

Probability

30 32 35

37

$$Pr(\mu-\sigma \leq x \leq \mu+\sigma) \approx 68\%$$

$$Pr(\mu-2\sigma \leq x \leq \mu+2\sigma) \approx 95\%$$

$$Pr(\mu-3\sigma \leq x \leq \mu+3\sigma) \approx 99.7\%$$